

ASRA

Reasoning, not memorization.

Adaptive Scientific Reasoning Architecture — a modular cognitive stack that spans research-grade flexibility, competition-ready agents, and closed-loop scientific discovery.

PLATFORM EVOLUTION

PyTorch-class research
+ Keras-class deploy
→ Scientific OS

Intelligence is not static prediction on fixed datasets. **ASRA behaves like a scientist** — observing, hypothesizing, experimenting, planning interventions, and refining world models under uncertainty.

STAGE 1 · RESEARCH STACK

Maximum flexibility. Composable cognitive phases, CLI eval harnesses, explicit epistemic objects.

asra-arc/src/asra/

STAGE 2 · COMPETITION LANE

Radical simplicity. Versioned Kaggle agents from asra-v0.4 through asra-v1.0 .

kaggle-notebooks/phase1-9

NINE-PHASE COGNITIVE ARCHITECTURE

Phase 1

Experience · transitions · state graphs

Phase 2

Observation · objects · rule hypotheses

Phase 3

Memory · exploration · subgoals

Phase 4

Causality · action semantics

Phase 5

Goal inference · hypothesis ranking

Phase 6

Planning · strategy invention

Phase 7

Robustness · generalization

Phase 8

Decision Biology bridge

Phase 9

Final integration · research story

9

Composable phases

2

Parallel lanes
research + deploy

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Unseen environments
ARC-AGI-3 stress test

Hypotheses as data structures

Transition-centric memory

Plan repair + stuck detection

SciLayer preprints

SCIENTIFIC INTELLIGENCE LOOP

- QuestionHidden objectives · ARC · biology
- ExperimentActions · perturbations
- ObserveTransitions · assays
- World modelState & pathway graphs
- HypothesizeGoals · rules · mechanisms
- PlanMulti-step interventions
- RefineMemory graph · abstractions

Decision Biology — the same architecture that adapts in ARC games reasons about cells: perturbations → actions, pathways → world models, adaptation → sequential decision-making under uncertainty.

EXPLORE ASRA

github.com/ilakkmanoharan/asra

SciLayer preprints

Kaggle Phase 7 agent

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Nature Foundation Models

Based on: From Databricks Meets PyTorch to Databricks Meets Keras
Industry white paper · June 2026